Deivaprakash K

Associate Robotics Engineer

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OBJECTIVE:

Robotics Engineer with hands-on experience building, testing, and deploying autonomous mobile robots using ROS2 and Python. Comfortable working across simulation, sensor integration, and on-robot validation in real environments. Looking to grow in a role that combines technical depth with real-world system integration.

EXPERIENCE:

GOAT Robotics, Coimbatore.
Associate Robotics Engineer | Jul 2024 – Present

- Led deployment and commissioning of **15+ autonomous mobile robots (AMRs)** across multiple Indian sites, including virtual deployments for **two international projects.**
- Reduced **on-site deployment time from 20 days to 7 days**, improving overall **deployment efficiency by 60%**.
- Supported **navigation, path planning, motion control, and pallet-handling development**, validating performance through simulation and on-robot testing.
- Acted as the **primary technical contact for clients**, conducting **35+ technical demonstrations** at customer sites and resolving live issues.
- Collaborated with hardware and software teams to **integrate sensors, controllers, and navigation stacks** for reliable robot operation.

My Equation, Ahmedabad, Gujarat.
Robotics & AI/ML Mentor | Apr 2025 – Present

- Mentored **400+ students and early-career engineers** in **robotics, AI, and machine learning**, including ROS2, navigation algorithms, and autonomous system workflows.
- Served as **one of the mentors for the ROS2 Nanodegree program**, guiding participants through hands-on projects and real-world robotics applications.
- **Played a major role in designing the syllabus** for the RoboAI program, contributing to course structure, project modules, and learning outcomes.
- Guided mentees on **robot simulation, mapping, and basic AI/ML model integration**, providing technical feedback and code reviews.
- Designed and conducted **workshops and training sessions**, helping learners understand robotics applications in industrial and research contexts.

Katomaran Technologies, Coimbatore
ROS Developer Intern | Sep 2023 – Jun 2024 (9 months)

- Used **Gazebo for simulation-based testing** and validation, comparing simulated behavior with real robot performance.
- Utilized **RViz2 and Foxglove** for visualization, debugging, and analysis of robot state, sensor data, and navigation performance.
- Performed real-world **testing of autonomous robots**, including **mapping, navigation, and docking workflows**, validating system behavior and identifying system-level issues.
- Developed **ROS2 health-check and diagnostic nodes** to monitor robot status, sensor availability, and system stability during testing and deployment.
- Worked with the **Nav2 navigation stack, supporting localization, path planning, controller tuning, and recovery behaviors** under real-world conditions.

TECHNICAL SKILLS:

- **Programming & Scripting:**
  - Python , Bash , Markdown (For technical documentation)
- **Robotics Frameworks:**
  - ROS2 (Robot Operating System), Nav2 (Navigation Stack),Path Planning, Control Systems , Moveit.
- **Simulation & Visualization Tools:**
  - Gazebo, Webots, Rviz, Foxglove , FlexSim Plant Simulation - for industrial flow and AMR visualization
- **Industrial & Collaborative Robotics:**
  - Hands-on exposure to collaborative robots (cobots) including FANUC, ABB,Techman, and Fairino.
- **Hardware Expertise:**
  - Extensive hands-on experience with AMR components such as Intel NUCs, 2D/3D LiDARs, Depth Cameras, IMUs, PLC's and Motor Drives.
- **AI / Machine Learning (Foundational):**
  - Strong foundation in supervised and unsupervised machine learning, with hands-on experience in Decision Trees and K-Means clustering, and familiarity with Reinforcement Learning concepts for autonomous systems and robotics.

CLIENT-FACING EXPERIENCE:

- Conducted **more than 35 technical demos showcasing AMR** capabilities to prospective clients with their applicational use cases.
- **Led on-site deployments of over 15 autonomous mobile robots** across various industrial environments in both Indian sites as well as Abroad sites.
- Actively involved in **client handling,** offering technical consultation, real-time issue resolution, and **ensuring successful integration.**
- Played a pivotal role in handover documentation, creating user guides, deployment reports, and acting as a **Customer Success Manager to bridge technical and client-facing operations**.

PROJECT & WORKFLOW MANAGEMENT:

- Proficient in ClickUp, GreytHR, OpenProject, Clockify, and Odoo for project tracking, CRM, and internal coordination.

ACADEMIC BACKGROUND:

EDUCATION	INSTITUTE	SCORE (CGPA)	YEAR
UG **(B.E Robotics and Automation)**	United Institute of Technology Coimbatore.	8.9 (Out of 10)	2020-2024 Batch

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ACHIEVEMENTS & HONORS:

- 1st Prize – e-Yantra Robotics Competition, IIT Bombay | 2024
- Cosmo Logistic Theme – Designed and implemented a robotics solution for material handling in a controlled warehouse environment, focusing on autonomous navigation, task execution, and system coordination.

PROJECTS:

Robotic Butler – Autonomous Navigation Simulation
Technologies: ROS2, Nav2, Gazebo, Simple Commander API

- Created a **custom Gazebo environment** and robot model, **simulating autonomous navigation using ROS2 Nav2.**
- Tested and validated **path planning, localization, and recovery behaviors** through repeated simulation runs.
- Additionally explored **Webots simulation** by implementing basic forward navigation for an **ePuck robot**, gaining exposure to multiple robotics simulation platforms.

3D LiDAR-Based Mapping (Large-Scale Outdoor Mapping)
Technologies: Livox MID360 LiDAR, Fast-LIO2, LIO-SLAM, ROS2

- Integrated Livox MID360 LiDAR with its inbuilt IMU for real-time **3D mapping.**
- Performed large-scale mapping (2+ km), validating system robustness in dynamic outdoor conditions.

Echo Navigator – Interactive Navigation Simulation
Technologies: ROS2, Nav2, Tkinter, RViz

- Developed a **simulated interactive robot with a GUI interface** for goal selection and navigation control.
- Used **RViz and ROS2 tools** to visualize robot state, sensor data, and navigation behavior.
- **Deployed in a controlled real environment** after simulation validation.

Simulation-to-Real Mapping & Localization Validation
Technologies: ROS2, RViz, Foxglove, **Fast-LIO2, LIO-SLAM**

- Validated localization and mapping performance by **comparing simulated navigation results with real-world LiDAR data.**
- Visualized and analyzed system behavior using **RViz and Foxglove.**

Smart Automated Vacuum Cleaner
Technologies: Arduino, C

- Built an **autonomous vacuum cleaner** with manual control capabilities, using a **Bluetooth module for remote operations** and C for programming logic.
